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23 December 1985
SUPERSEDING
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17 August 1979

MILITARY SPECIFICATION

CLEANER, PRESSURE, WATER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of stationary, electric motor driven pressure cleaner, equipped with a pressure pump, hoses, and manually operated nozzles, for cleaning motor vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.
ZZ-H-561	-	Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-C-843 - Cushioning Material, Cellulosic.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape, Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services.
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02	-	Manual for Machine Tool Gearing.
AGMA 390.03	-	Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82	-	Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54	-	Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1	-	Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1	-	General Standards for Industrial Control and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	-	National Electrical Code.
NFPA 79	-	Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These

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documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The pressure cleaner shall be new and of the manufacturer's latest design that shall be capable of meeting the requirements specified herein. The basic design shall consist of an electic motor driven water pump, a base, pressure hoses, a cleaning wand, and nozzles. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair without modification, distortion, or damage to the cleaner or any cleaner part or component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the cleaner. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

3.2.2 Gears. All gears and pinion designs selected for use in the cleaner or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and FED-STD-376.

3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.4 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.5 Maintainability. Design of the cleaner, and the placement of adjusting features, shall assure that maintenance of operational capability will be reduced to the simplest form of procedures requiring a minimum of skill. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

3.2.6 Safety and health requirements. The pressure cleaner shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the cleaner itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

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3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The cleaner shall be constructed so that it is capable of meeting all the requirements specified herein and, when installed and connected to the specified water and power source, can perform all operations for which intended.

3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hardspots, shrinkage defects, cracks, and other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the cleaner to remove the temporary caps or stoppers before start-up or use of the cleaner.

3.4.5 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.4.6 Electrical components. Electrical components, including relays, starters, switches wiring, and their components, shall conform to, and be located in accordance with, NEMA ICS 1 and NFPA 70, as applicable.

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3.4.7 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

3.4.8 Electrical wiring. Wiring shall be in accordance with NFPA 79. Provision shall be made to assemble and wire all components in a way that assures adequate equipment grounding. Liquid-tight, flexible connections shall be made between controls and motor.

3.4.9 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.4.10 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.5 Components. The pressure cleaner shall consist of, but not be limited to, the principal components covered in 3.5.1 thru 3.5.11.2.

3.5.1 Base. A metal base shall be provided, capable of supporting the entire pump unit, power drive, motor, and pressure gage. Handles shall be provided on the base to facilitate lifting the washer by the operator. When specified (see 6.2.1), a wheeled cart shall be provided to allow for cleaner transportability.

3.5.2 Pump. The pump shall deliver not less than four gallons of water per minute (gpm) at a steady rate at an operating pressure of 600 pounds per square inch gage (psig) for a period of 40 hours (five eight hour periods). Two water outlets shall be provided, each controlled by a valve. The outlets shall operate individually and simultaneously to allow for use of one or two cleaning wands. The pump cylinder walls and plungers shall be made of corrosion resistant metal. The pump shall be provided with means for lubrication. The pump unit and all components shall meet all performance and endurance requirements specified herein when operated using water ranging in temperature from + 32° to +140°F.

3.5.2.1 Power drive. Power transmission from the motor to the pump may be through pulleys and V-belts, direct drive, silent chain drive, a gear reduction unit, or by use of a gear motor. Except for the V-belt type, the drive shall be totally enclosed. When a V-belt drive is used, means shall be provided to allow adjustment for proper belt tension. All rotating shafts shall be mounted on ball or roller bearings. The silent chain drive, if used, shall be self-oiling by an immersion or splash system.

3.5.2.2 Valves. The seating surfaces of the pump valves and valve seats shall be corrosion resistant and replaceable.

3.5.2.3 Stuffing box(s). Plunger type pumps shall be provided with stuffing (packing) box(s) around the plunger rods. The stuffing box(s) shall permit adjustment and replacement of packing material, if applicable.

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3.5.3 Intake lines. Two intake lines shall provide water to the pump. One intake line shall provide connection to the suction hose (see 3.5.5). The second intake line shall be a threaded, 3/4 inch, quick-disconnect type.

3.5.4 Chemical injection. Means shall be provided to inject liquid chemicals (soap) into the water flow.

3.5.5 Suction hose. A suction hose equipped with a suitable strainer shall be provided. The hose shall be not less than six feet long and shall conform to the requirements for grade A, type I, one inch inside diameter, specified in ZZ-H-561. The hose shall be provided with a quick-disconnect coupling suitable for interfacing with the intake line inlet.

3.5.6 Pressure hoses. Two pressure hoses, 1/2 inch inside diameter by 35 feet in length, shall be provided. The hoses shall be of reinforced rubber construction and shall withstand an operating pressure of not less than 750 psig. The hose, with couplings, shall have a bursting pressure of not less than 2250 psig. The couplings shall allow for proper interfacing of the pressure hoses to the cleaning wands and the intake line inlets.

3.5.7 Cleaning wands. Two cleaning wands, designed for vehicle cleaning, shall be provided. The wand handles shall be provided with on-off valves. The wands shall be designed for use with the nozzles specified in 3.5.8.

3.5.8 Nozzles. Two straight stream nozzles and two fan type nozzles shall be provided. The nozzles shall be fabricated of corrosion resistant steel and shall each deliver at least two gpm at a pressure of 600 psig.

3.5.9 Pressure gage. Water pressure shall be indicated by a dial pressure gage. The gage shall indicate pressure over a range of at least 0 to 750 psi. The gage shall be graduated in increments of not more than 25 psi and numbered at intervals of not more than 100 psi.

3.5.10 Pressure regulating governor. An adjustable, pressure regulating governor shall be provided to maintain an operating pressure of at least 600 psig, with a by-pass valve to permit release of discharge pressure back to the suction side when either or both nozzles are closed.

3.5.11 Motor. The motor shall be an induction squirrel-cage type and shall operate from 230/460 volts, three phase, 60 Hertz. The motor shall conform to the requirements of CC-M-1807, with class B insulation and a continuous duty rating, and be initially wired for 230 volts.

3.5.11.1 Motor switch. A weatherproof, manually operated, "on-off" switch with overload protection shall be provided on the motor. The switch shall be accessible from the operator's side of the cleaner.

3.5.11.2 Power cable. A power cable at least ten feet in length shall be properly wired to supply power to the motor. The power cable shall have a voltage and current carrying capacity equal to, or more than, that required by the motor and NFPA 70. The cable shall have water and grease resistant insulation, type SO, conforming to NFPA 70. The terminal end shall be stripped one inch and tinned.

3.6 Product identification. The cleaner shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

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3.7 Workmanship. Standards of workmanship shall assure that the pressure cleaner shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

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Examination/Inspection	Inspection Level	AQL
Product examination (4.5)	II	1.0
Performance test (4.6.1)	S-4	1.0
Packaging inspection (4.7)	S-4	4.0

4.5 Examination. Visually and manually examine the cleaner to determine conformance with the requirements of 3.2 thru 3.4.10, 3.5.1, 3.6, and 3.7. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of moveable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the cleaner and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. The pressure cleaner shall be connected and operated in accordance with the manufacturer's operating instructions. With both nozzles open, the pump shall be operated at the minimum discharge rate of four gpm at not less than 600 psig operating pressure for a period of 30 minutes. During this 30 minute period, the pump shall be operated continuously; however, after each seven minutes of "open" operation, both nozzles shall be completely closed for a period of at least three minutes in order to check the operation of the by-pass valve and pressure regulating governor. (Inlet water temperatures for this test shall be at least 130°F.) Upon completion of this test, there shall be no evidence of failure or damage to the pressure cleaner or any of its component parts. There shall also be no change in operating pressure or discharge rate. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component shall constitute failure of this test.

4.6.2 Capacity test. The pressure cleaner shall be operated at not less than 600 psig. A water meter shall be attached to the discharge outlet or the inlet of the pump to visually indicate the delivery of four gpm total to the two nozzles operating at the same time (see 3.5.8). Both nozzles shall be closed for a period of three minutes to establish that the pressure regulating governor and the by-pass valves function properly. When the nozzles are again opened, any change noted in either the delivery rate or operating pressure with respect to the original test conditions shall constitute failure of this test.

4.6.3 Endurance test. With both nozzles open, the pump shall be operated at an operating pressure of not less than 600 psig for a period of 40 hours (five eight hour periods). The inlet water temperature shall be 100° ±10°F. During the eight hour periods, the pump shall be operated continuously; however, after each 27 minutes of "open" operation both nozzles shall be completely closed for a period of three minutes in order to check the operation of the by-pass valve and pressure regulating governor. Upon completion of this test, there shall be no damage to the pressure cleaner or any of its component parts. The pressure cleaner shall be operated at its normal settings and there shall be no change in operating pressure or discharge rate. Stuffing boxes, hose connections, and control valves shall show no evidence of water leakage. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component shall constitute failure of this test.

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4.6.4 Pressure hose test. A sample hose shall be subjected to a hydrostatic test pressure of 2250 psig. The sample shall not burst or show evidence of leakage after two minutes of test.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. The pressure cleaner may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The complete pressure cleaner assembly shall be cleaned and dried in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with process C-1 and drying shall be in accordance with any applicable procedure specified therein.

5.2.1.2 Preservative application. Preservatives and their application shall be in accordance with MIL-P-116. Immediately after cleaning and drying, all unpainted exterior ferrous metal surfaces of the pressure cleaner, including those exposed by disassembly, shall be coated with type P-1 preservative. Interior surfaces of the pump, water system, and crankcase shall be coated with type P-10, grade 30, preservative. Apply by flushing or pumping the preservative through the entire system. Excess preservative shall be drained. Small openings such as drain cocks, shut-off valves, and pipe connections shall be sealed with tape conforming to PPP-T-60, type optional, or caps or plugs conforming to MIL-C-5501.

5.2.1.3 Basic unit. The basic pressure cleaner shall be individually preserved in accordance with method I of MIL-P-116.

5.2.1.4 Gages. Gages disassembled from the basic unit shall be preserved by sealing all openings with tape conforming to PPP-T-60, type IV. The gages shall be packed in a snug-fitting box conforming to PPP-B-566, variety 2. All gages, including those attached to the basic unit, shall be protected with cushioning material conforming to PPP-C-843, type II. Secure the cushioning material with tape conforming to PPP-T-60, type optional.

5.2.1.5 Cleaning wands and nozzles. Cleaning wands and nozzles shall be preserved with type P-10, grade 30, preservative and packed method I in accordance with MIL-P-116.

5.2.1.6 Hoses and fittings. Hoses and fittings shall be packed in accordance with MIL-P-775.

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5.2.1.7 Electrical equipment. The electric motors shall be preserved in accordance level A requirements of MIL-E-16298, using the alternate method specified for assembled machines. Exposed ends of wire, plug openings, connector plugs, and openings into switch boxes shall be sealed with tape conforming to PPP-T-60, type optional. Coil the electric cable to a minimum safe diameter and secure with cotton tape or twine to prevent uncoiling.

5.2.1.8 Drive belts. Drive belts shall be removed from the equipment or released from tension. The face or grooves of all ferrous metal pulleys shall be coated with primer coating conforming to TT-P-636. Belts removed from the equipment shall be packed in accordance with method III of MIL-P-116.

5.2.1.9 Technical data. When furnished, publications issued with the cleaner shall be packed in accordance with method IC-1 of MIL-P-116.

5.2.1.10 Maintenance tools. Maintenance tools issued with the pressure cleaner shall be preserved and packed in accordance with PPP-P-40.

5.2.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

5.3 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

5.3.1 Level A. The complete cleaner shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The arrangement of contents, blocking, bracing, anchoring, and cushioning shall be in accordance with the container specification and MIL-STD-1186.

5.3.2 Level B. The cleaner shall be packed as specified in 5.3.1 except that the shipping container shall conform to PPP-B-601, domestic type, or PPP-B-621, class 1.

5.3.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.4 Marking. Marking shall be Military or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military marking is required, it shall be so specified in the contract.

5.4.1 Military marking. Military marking shall be as specified in MIL-STD-129.

5.4.2 Industrial marking. Industrial marking shall be as specified as in ASTM D3951-82.

6. NOTES.

6.1 Intended use. The pressure cleaner is intended for use in base shops, field service units, and maintenance areas for washing vehicles.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.6).
(d) Painting and finishing if different (see 3.4.10).
(e) Wheeled cart when required (see 3.5.1).
(f) Product identification if different (see 3.6).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Sampling if different (see 4.4.1).
(j) Level of preservation, packing, and marking if different (see 5.2, 5.3, and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Standard lubricants. The cleaner manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - YD

Review activity:
Army - SM

User activity:
Navy - MC

Preparing activity:

Army - AL

Project No. 4940-0480

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